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GROUP 39

TITLE:

**PREDICTING A MOVIE RECOMMENDATION SYSTEMS ON NETFLIX USING
SUPERVISED LEARNING TECHNIQUES AND SENTIMENT ANALYSIS**

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1.0 INTRODUCTION

Redditor Reed Hastings and Marc Randolph launched the popular streaming service Netflix in 1997, and it has completely transformed the entertainment sector. With more than 200 million members globally, Netflix has established itself as a major force in the distribution of digital material. The platform's creative application of recommendation algorithms and data analytics, which offers viewers personalized content recommendations and improves their viewing experience, has been essential to its success.

Two datasets were obtained via Kaggle, which are titled "Netflix IMDB Scores," offers an extensive compilation of Internet Movie Database (IMDb) ratings for Netflix productions, and "Sentiment Analysis of IMDB Movie Reviews," contains user reviews from IMDb. By integrating these datasets, it provides a comprehensive understanding of audience perceptions and content performance through the analysis of both quantitative ratings and qualitative reviews.

In the film and television industry, understanding user preferences and content success are critical in today's competitive entertainment market. However, it is still lacking determining whether a movie or show is recommended to watch or not recommended to watch based on some important attributes. There are too many movies and shows available on Netflix and without a movie recommendation system, it is difficult to make data-driven decisions about marketing tactics, and content acquisition, which will eventually increase audience engagement and pleasure. In addition to movie recommendation system, understanding user sentiments are needed to improve Netflix in the future. However, Netflix might not focus on the users' comment, and it might cause users' dissatisfaction. It is still lacking on extract the knowledge from comments on Netflix.

Thus, to address these business issues and obtain practical insights that will improve audience satisfaction and content strategy, the main objectives of the study are:

1. To predict which predictive modelling techniques perform well in movie recommendation system.
2. To understand the main keywords and responses on Netflix by applying sentiment analysis

2.0 RELATED WORK

First, predictive analytics techniques can be applied for movie or show recommendation system. Khan et al. (2020) would like to discover viewing patterns of users who are using movie applications, such as Netflix. Therefore, they proposed Naive Bayes classifiers and integrated with various feature sets, such as unigrams, bigrams, and trigrams to identify the viewing patterns. As a result, the accuracy of all Naive Bayes with various features are above 85% using IMDb datasets that contain 50,000 reviews. Although there is a high accuracy, both average precision and recall are between 0.40 and 0.50.

Another study also conducted predictive analytics study that examined the performance of sentiment analysis on the movie reviews by applying decision trees, random forest, gradient boosting classifier and support vector machines (SVM) (Naeem et al., 2022). Their study's focus is classifying the reviews into positive and negative reviews. As a result, they discover SVM leads to the highest accuracy of all selected features compared to other predictive modelling techniques.

Second, sentiment analysis was also used to understand the sentiment of all movie reviews. Dalkılıç and Çam (2021) would like to identify the sentiment of tweets about movie reviews in Turkish by selecting particular hashtags. As a result, among 19,398 tweets that imported from Twitter, they discovered more than 50% of the tweets are negative sentiments.

Thus, according to related works above, this study will fill the literature gaps. First, this study will apply K-Nearest Neighbor (KNN) in the predictive modelling section as it has not been used to classify the movie recommendations system. In order to combine the result of accuracy using KNN, SVM technique also will be applied in the predictive modelling section. Second, for the sentiment analysis, this study will examine the overall sentiment of all movie reviews and identify the keywords from all positive and negative reviews.

3.0 DATA EXPLORATION

Netflix IMDB Scores dataset provides a comprehensive overview of various titles available on the popular streaming platform, Netflix. This dataset is ideal for analyzing patterns, trends, and user preferences related to Netflix content. Here's a detailed breakdown of the dataset:

- **Source:** The dataset is derived from Netflix's streaming catalog, enriched with IMDb ratings and votes, and includes metadata from Kaggle. <https://www.kaggle.com/datasets/thedevastator/netflix-imdb-scores>
- **Size:** The dataset consists of 5282 entries, each representing a unique title available on Netflix. There are 10 columns in this dataset.
- **Attributes:**
 - **id:** The unique identifier for each entry.
 - **title:** The name of the TV show or movie.
 - **type:** Indicates whether the entry is a TV show or a movie.
 - **description:** A brief summary of the plot or storyline.
 - **release_year:** The year the TV show or movie was released.
 - **age_certification:** The age rating (e.g., G, PG, R) indicating suitability for different audiences.
 - **runtime:** The length of episodes for TV shows or the duration of movies.
 - **imdb_id:** The unique identifier for IMDb.
 - **imdb_score:** The IMDb rating score of the title.
 - **imdb_votes:** The number of votes the title received on IMDb.

In addition to Netflix IMDB Scores dataset, IMDB Dataset of 50K Movie Reviews dataset was also used to conduct sentiment analysis. Here's a detailed breakdown of the dataset:

- **Source:** The dataset is derived from Kaggle.
<https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>
- **Size:** The dataset consists of 49582 unique entries and 2 columns, consisting of review and sentiment.
- **Attributes:**
 - **review:** The comment for particular movies
 - **Sentiment:** Pre-defined sentiment of review, which are positive, negative or neutral. This study will not apply this attribute for sentiment analysis.

3.1 DATA CLEANING AND PRE-PROCESSING

To ensure the dataset is ready for analysis, several data cleaning steps are performed:

1. Handling Missing Values:

- a. **age_certification:** Replace null values with "G" (General Audiences) to maintain consistency and prevent null entries.
- b. **imdb_votes:** Fill missing values with the mean number of votes, ensuring that the dataset reflects a realistic average rather than misleading zeros.

2. Column Adjustments:

- a. **description:** This column can be dropped since it is not needed and irrelevant for analysis that we are going to do.
- b. **imdb_score:** Convert scores to one decimal place for uniformity.

3. Creating Output Labels for Predictions:

- a. Titles with an IMDb score greater than 7 are labeled as "yes" in a new column named "recommended" indicating they are generally well-received. Titles with a score of 7 or below are labeled "no" indicating they are less recommended.

Overall, the Netflix TV Shows and Movies dataset is a rich resource for analyzing various aspects of content available on Netflix. By cleaning and preparing the data, it can help employing effective visualization techniques, so that we can uncover valuable insights into viewer preferences and content trends on the platform.

4.0 DESCRIPTIVE ANALYTICS AND DATA VISUALISATION

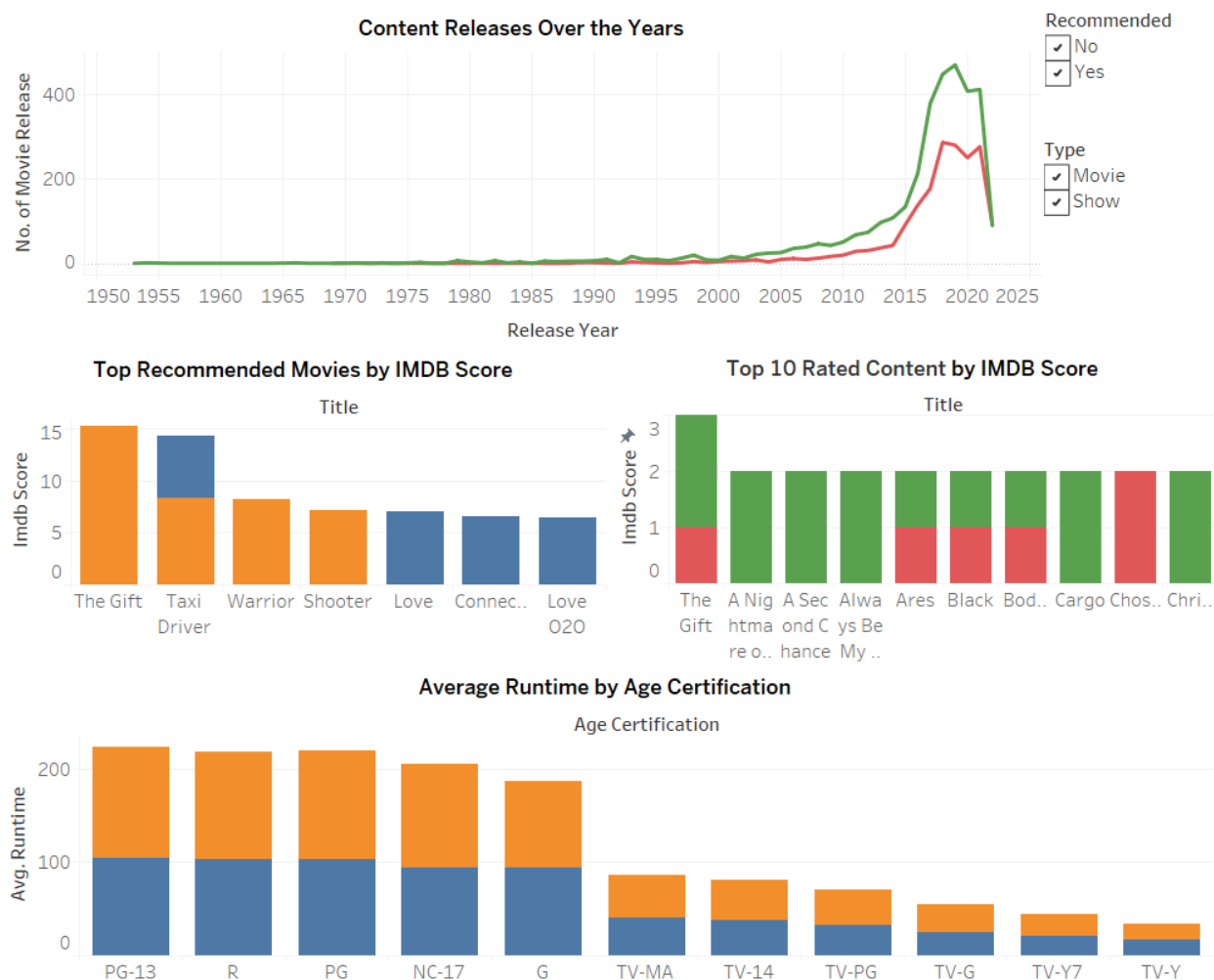


Figure 1: Dashboard for Netflix IMDb scores (Link: [Tableau Public Link G39](#))

The first content releases over the years graph shows the number of movies and shows released each year from 1950 to 2025. The green line represents the movie, while the red line represents the show. There is a significant increase in content releases from around 2000 to around 2020. This upward trend reflects the growing entertainment industry, likely due to the rise of streaming platforms and increased production budgets. The decrease starting 2020 could be a result of external influences like the COVID-19 outbreak. In 2023-2024, a drastic reduction in the releases of content, especially the show type, could be caused by factors of the pandemic and many restrictions to cure the outbreak through movement control order (MCO).

The bar chart at the left side displays the IMDb scores of the top recommended movies, with orange representing recommended movies and blue representing non-recommended movies. The movies "The Gift" and "Taxi Driver" stand out with the highest IMDb scores, followed by "Warrior" and "Shooter" which have moderate scores. The other movies have relatively lower

scores. Non-recommended movies like "Connection" and "Love O2O" have lower scores. The top 4 movies are also the most recommended by others.

The bar chart at the right side shows the IMDb scores of the top 10 rated content, including both movies and shows. "The Gift" again appears with the highest score, followed by several other titles with lower but relatively similar scores. The repeated appearance of "The Gift" indicates its strong popularity and high rating. The mixed scores for other content suggest a diversity in audience preferences and the types of content that receive high ratings.

The bar chart at the bottom represents the average runtime of content based on different age certifications. Content rated PG-13 (Parental Guidance-13) and R (Restricted) have the highest average runtimes, followed closely by other age certifications that generally display lower average runtimes. Content with age certifications like PG-13 (Parental Guidance-13) and R (Restricted) tend to have longer runtimes, possibly because these categories include many feature films. TV shows, which often have multiple episodes, tend to have shorter average runtimes per episode. This reflects the different formats and consumption patterns, with movies often being longer in duration compared to episode TV content.

In summary, the rise in content releases over the years reflects the growth of the entertainment industry, particularly with the rise of streaming services like Netflix. The diversity in top recommended and top-rated content will help effective recommendation systems and varying audience preferences. Longer runtimes for certain age certifications align with will help the production of feature-length content reflecting different consumption patterns. In this analysis, the key insights provided about the trends in content production, audience preferences, and viewing habits, which are valuable for content creators, marketers, and streaming service providers.

5.0 DATA ANALYSIS

In this section, two analyses will be performed, which are predictive modelling and sentiment analysis by using RapidMiner.

5.1 PREDICTIVE MODELLING

Two predictive analysis techniques were used to predict and classify the recommendation of a movie and show. There are support vector machines (SVM) and K-nearest neighbors (KNN).

Figure 2 illustrates the process of the SVM and KNN using RapidMiner, and Figure 3 illustrates the process of cross validation for support vector machine technique.

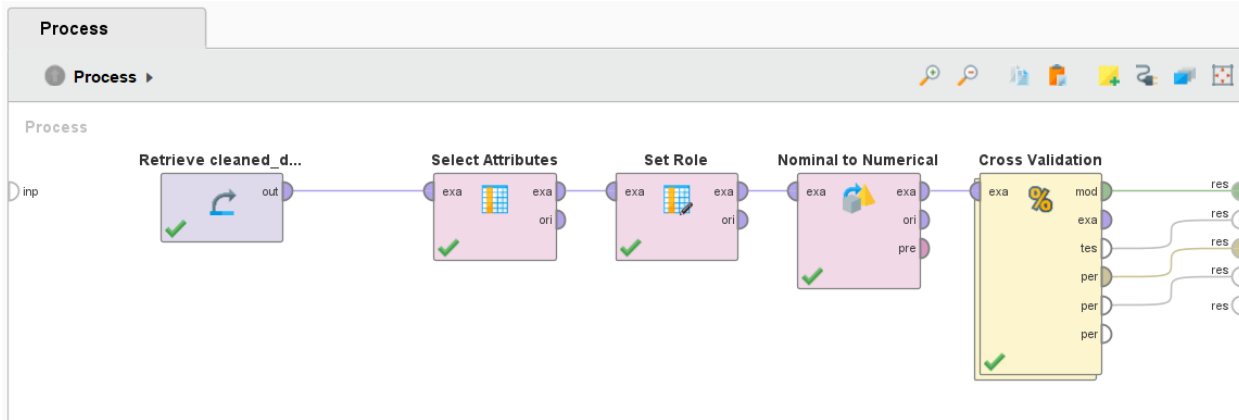


Figure 2: The process of the SVM and KNN

After retrieving the cleaned data, to conduct predictive analysis, only several attributes are selected, which are “type”, “age_certification”, “runtime”, “imdb_votes”, and “recommended”. In the “Set Role” operator, “recommended” was selected as label and other attributes are features. Since the dataset contains nominal or text data, it is important to convert the nominal data to numerical, so that it can be recognised by machine. 10-fold cross-validation was used for data splitting.

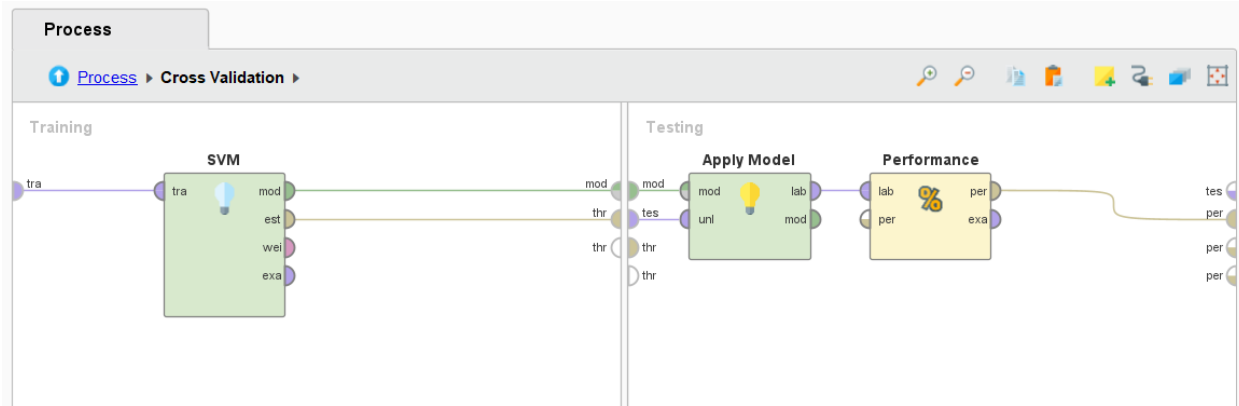


Figure 3: The cross-validation process of the SVM

In the cross-validation process, “SVM” operator was applied to train the model. To test the accuracy, “Apply Model” and “Performance” operator were selected. As a result, confusion matrix and accuracy will display after executing all processes. Figure 4 displays the result.

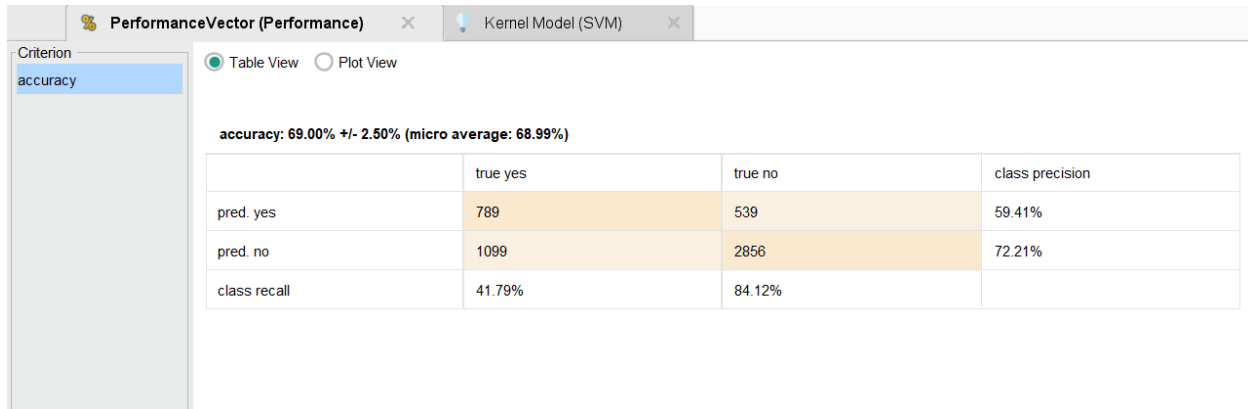


Figure 4: The results of accuracy and confusion matrix using SVM technique

Based on Figure 4, the accuracy of Netflix recommendations system is 69.00%. In the confusion matrix table, True Positive (TP) is 789 units, False Negative (FN) is 2856 units, True Negative (TN) is 1099 units, and False Positive (FP) is 539 units. The precision records 59.41% and recall records 41.79%.

Besides SVM technique, KNN technique also applied in the predictive analysis. The process is like SVM except there is a different part in cross-validation process (see Figure 5). Then, Figure 6 shows the results of accuracy and confusion matrix using KNN technique.

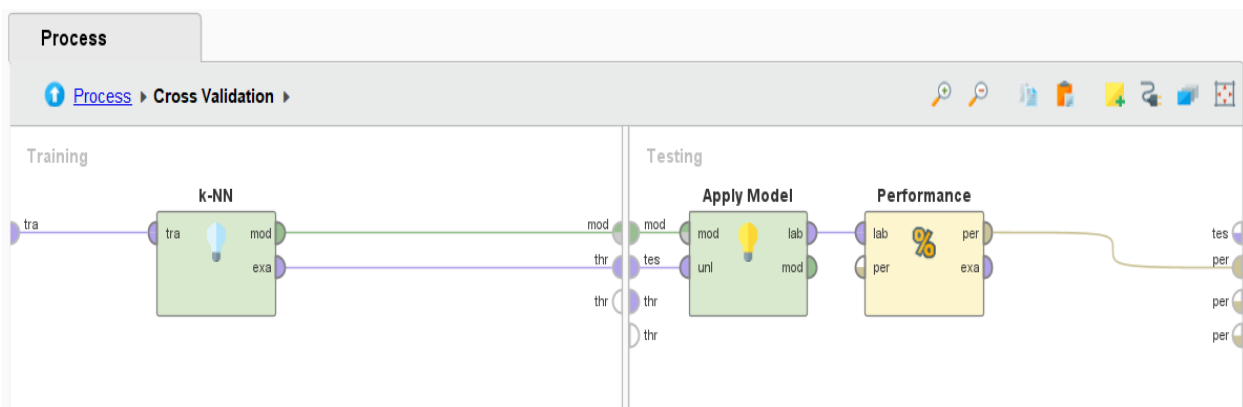


Figure 5: The cross-validation process of the KNN

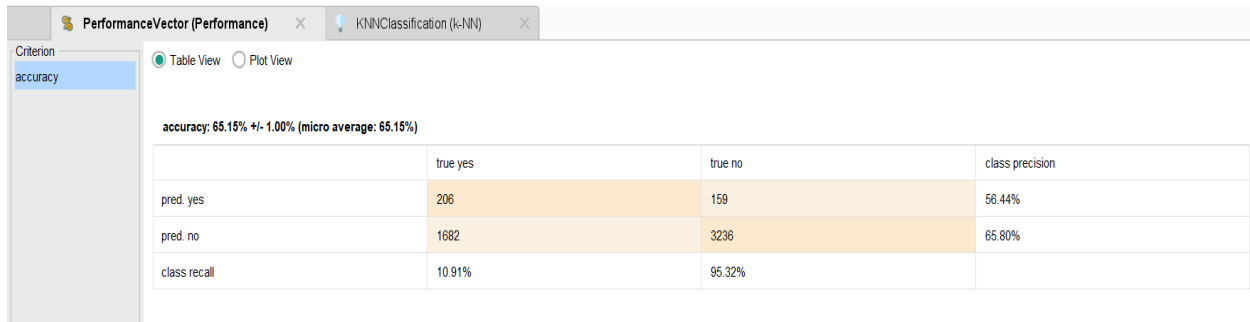


Figure 6: The results of accuracy and confusion matrix using KNN technique

Based on Figure 6, the accuracy of Netflix recommendations system is 65.15%. In the confusion matrix table, True Positive (TP) is 206 units, False Negative (FN) is 159 units, True Negative (TN) is 1682 units, and False Positive (FP) is 3236 units. The precision records 56.44% and recall records 10.91%.

5.2 SENTIMENT ANALYSIS

In addition to recommend a good movie, it is crucial to understand the reaction from audiences about a movie or show. A dataset named “IMDB dataset of 50k movie reviews” was used to conduct sentiment analysis using RapidMiner. Figure 7 below illustrates the process of sentiment analysis in RapidMiner.

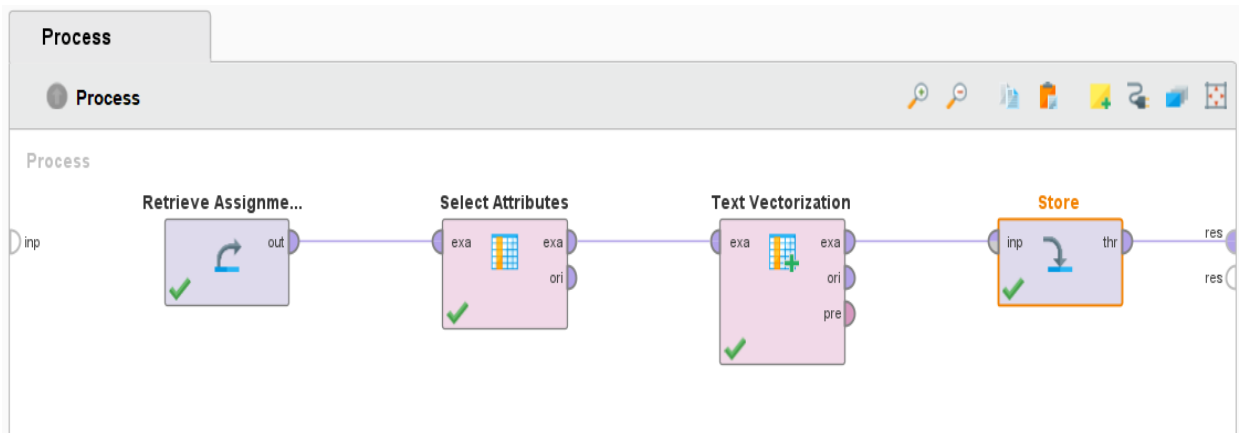


Figure 7: The process of sentiment analysis

After retrieving the dataset that was used for sentiment analysis, “review” attribute was used to conduct sentiment analysis. “Text Vectorization” operator performs the transformation from texts into a vectorised format. After that, the results will be stored in RapidMiner, and it will perform for the next step.

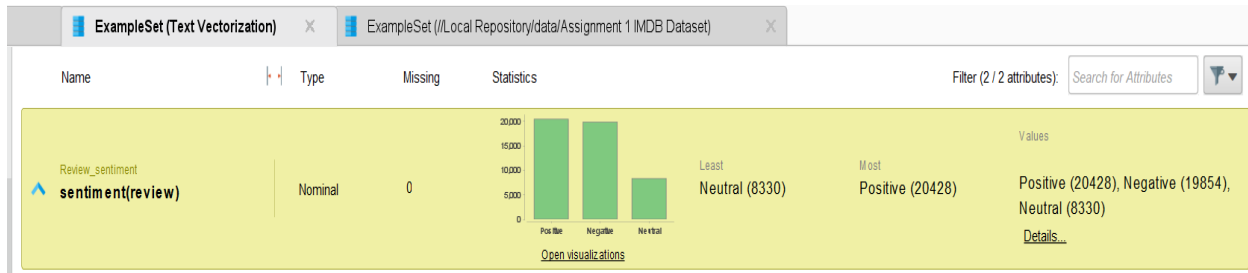


Figure 8: Statistics of the sentiment analysis for all reviews

According to Figure 8, among 48612 unique reviews, most of the reviews are positive, which record the highest number (20428 reviews, 42.0%), followed by the negative reviews (19854 reviews, 40.8%), and 8300 reviews (17.1%) are categorised neutral. Figure 9 below illustrates the pie chart of sentiments analysis for all reviews in RapidMiner.

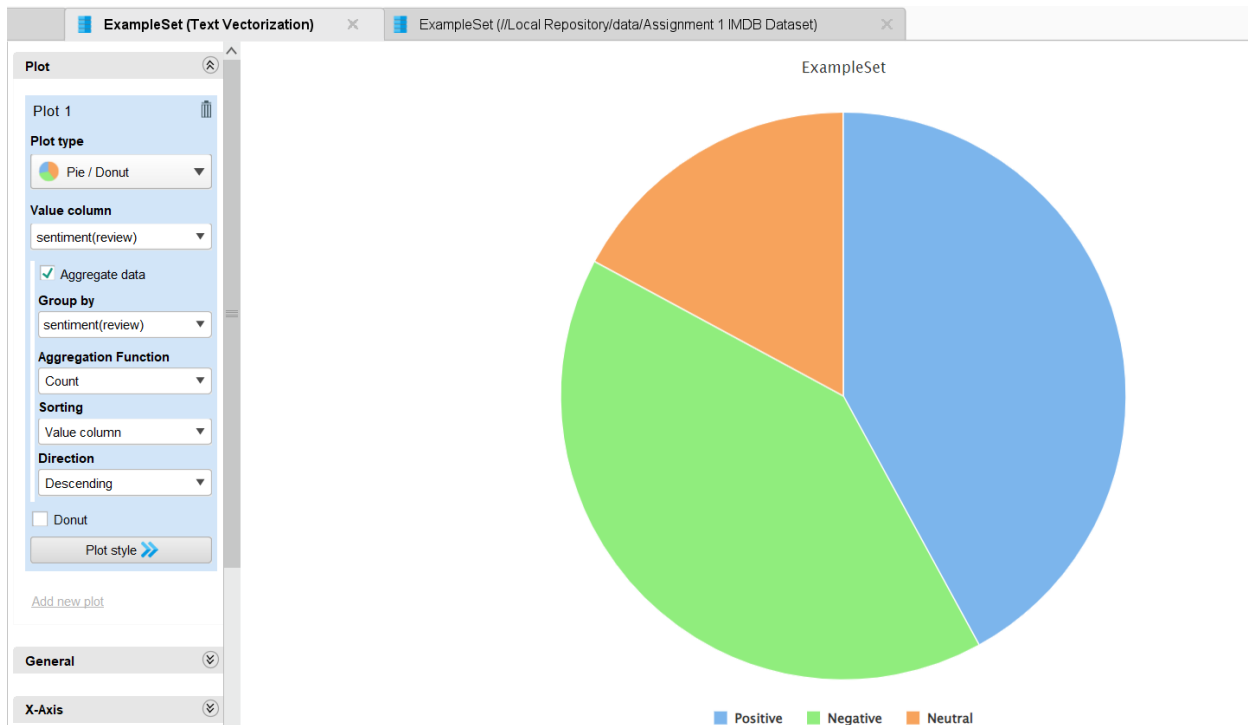


Figure 9: Pie chart of sentiment analysis for all reviews

Next, in order to identify the keywords from the positive and negative reviews from datasets. A word cloud was applied in this sentiment analysis to visualise the keywords in a different size based on total occurrence in all reviews. Figure 10 illustrates the process of display total occurrence for each word in RapidMiner.

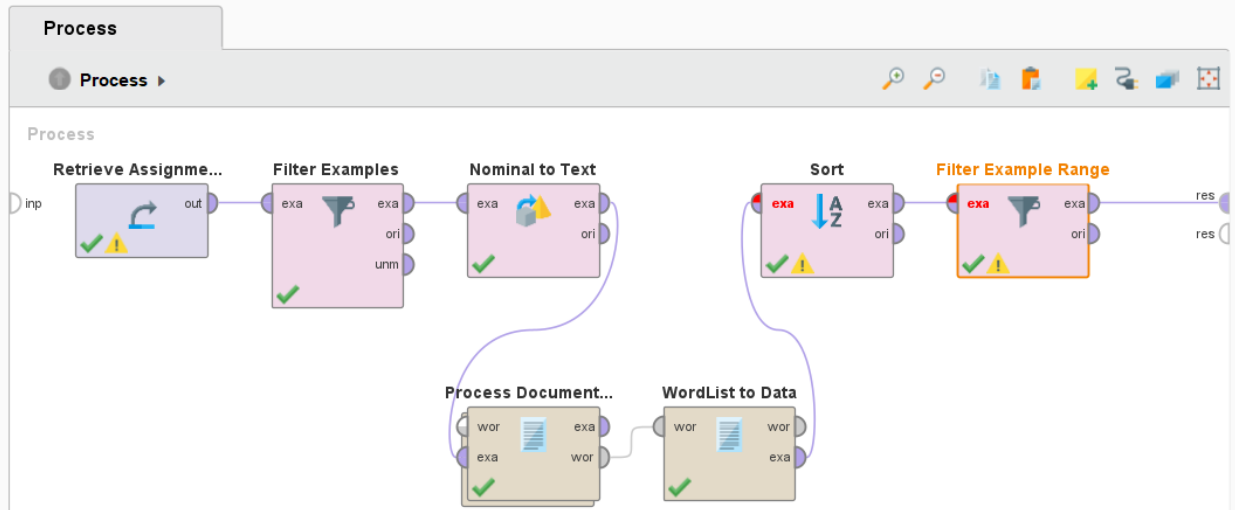


Figure 10: The process of display total occurrence for each word in RapidMiner

According to Figure 10, the data retrieved from the result of sentiment. In the “Filter Examples” operator, only the reviews that recognised positive reviews will be used to display word cloud. The next step is transforming the nominal data to text data. Process documents from data is a crucial step in text processing analysis. Figure 11 illustrates the process documents from data in RapidMiner.

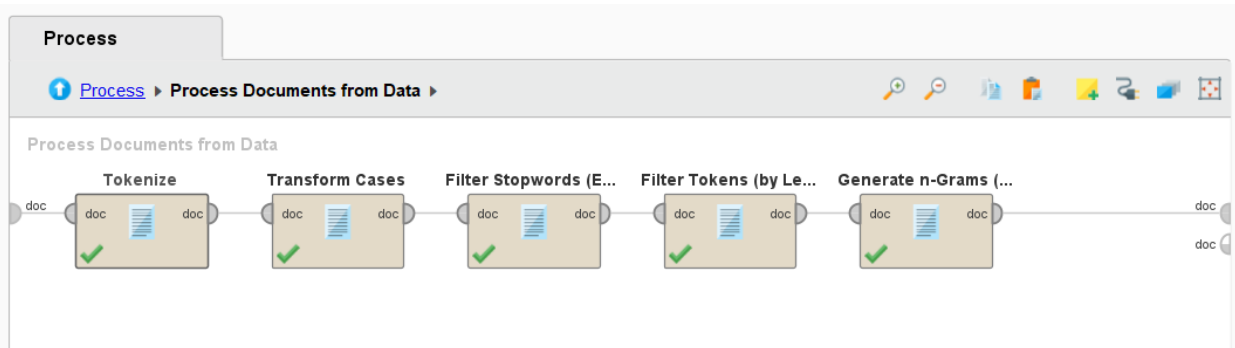


Figure 11: Process documents from data

“Tokenize” operator executes splitting a word from a review into individual words, “Transform Cases” operator executes converting all text from all reviews into a lower-case format, “Filtering Stopwords (English)” operator executes removing all common words that are less likely contribute to overall analysis, “Filter Tokens (by Length)” operator executes removing word or token that is less than 4 letters and more than 15 letters, and “generate n-Grams (Terms)” operator executes allowing 1-grams or known as individual words contribute in overall analysis.

After that, “WordList to Data” operator is selected to calculate the occurrence of each word and transform it into data. By selecting “Sort” and “Filter Example Range” operators, only the top 100

words that are the most frequent words will be displayed in the result (see Figure 12) and the result will be transformed to word cloud (Figure 13).

Result History

ExampleSet (Filter Example Range)

Open in

Turbo Prep

Auto Model

Interactive Analysis

Filter (100 / 100 examples): all

Data

Statistics

Visualizations

Annotations

Row No.	word	in documents	total
1	film	11494	33761
2	movie	11316	29855
3	good	7318	11423
4	story	6738	10658
5	great	6731	10261
6	time	6920	10093
7	love	4628	7048
8	life	4353	6590
9	people	4512	6503
10	films	4004	6135
11	characters	4155	5809
12	made	4560	5752
13	character	3827	5493
14	movies	3920	5481
15	think	4006	5395
16	seen	4337	5380
17	watch	4113	5251
18	show	2767	5230

ExampleSet (100 examples, 0 special attributes, 3 regular attributes)

Figure 12: The result that shows only the top 100 words



Figure 13: Word cloud (positive words)

Other than positive review, negative reviews was also examined and display in a tabular format and word cloud. Figure 14 shows the results that shows only the top 100 words in negative reviews and Figure 15 illustrates its word cloud.

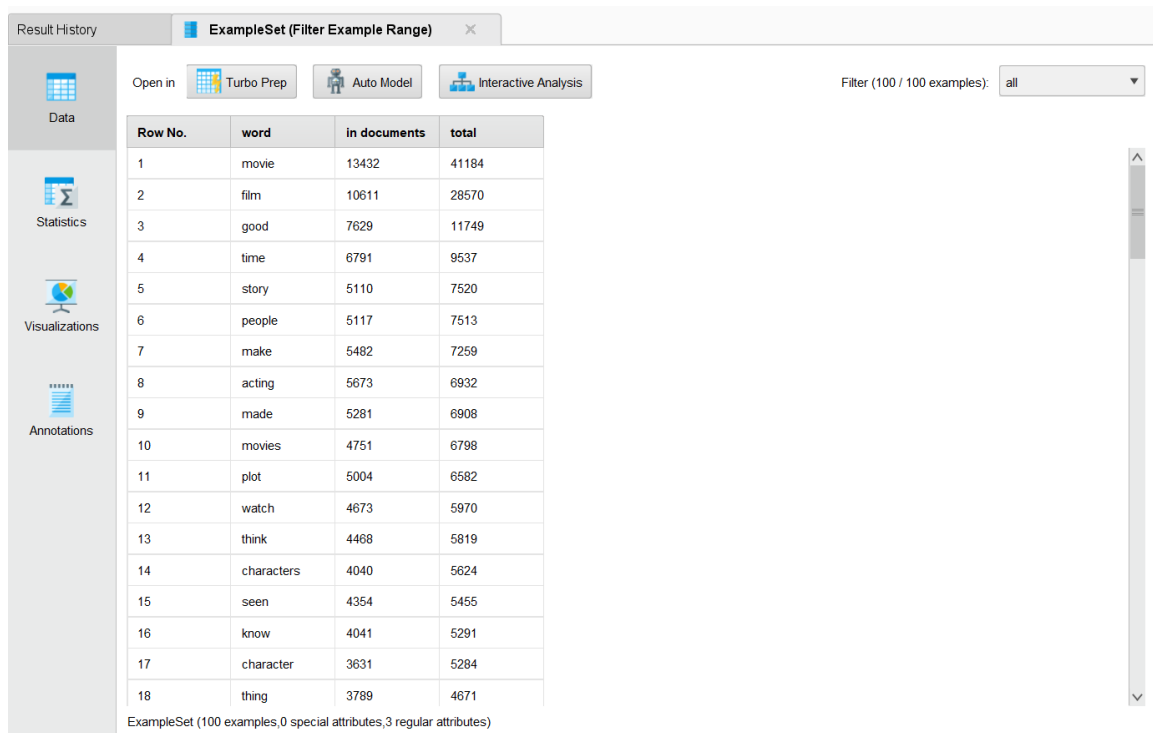


Figure 14: The result that shows only the top 100 words

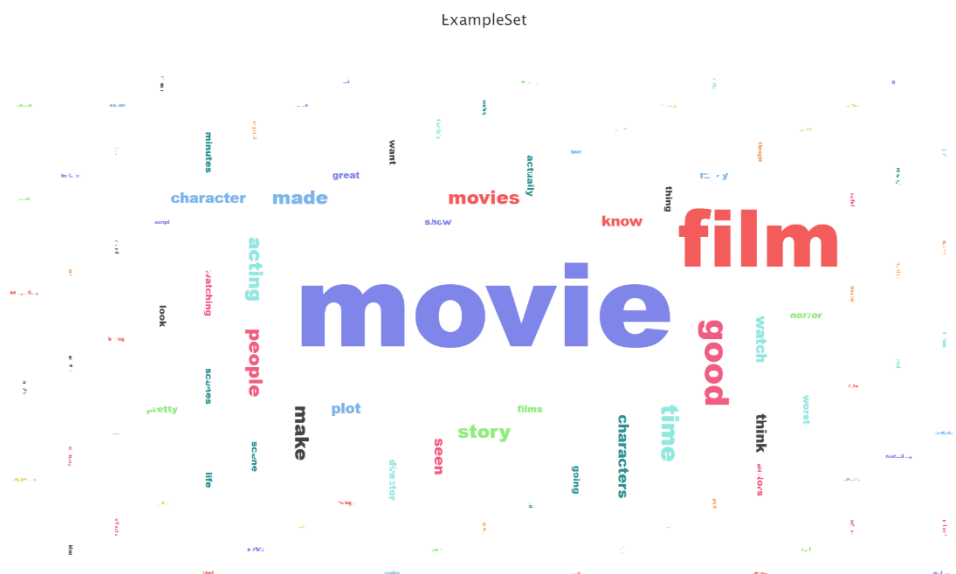


Figure 15: Word cloud (negative)

6.0 DISCUSSION AND CONCLUSION

For the predictive modelling section, it is clearly to see that the accuracy of SVM (69.00%) greater than KNN (65.15%). It indicates that SVM technique works a better performance in the movie recommendation system on Netflix. In fact, this result is consistent with a study's outcome that mentioned at related work section (Naeem et al., 2022). Thus, it can be used to predict whether a movie is recommended to watch or less likely recommended to watch on Netflix.

Next, interestingly, the precision and recall under SVM technique are quite low, which are 59.41% and 41.79% respectively. It is because the higher the result of precision, the higher the prediction being marked correctly. Meanwhile, the higher the result of recall, the greater the model correctly identifies TP. Eventually, the probability of incorrectly recommending a movie might be high. To improve the results of precision and recall, more attributes are needed in the datasets, such as movie types. Although the number of TP in SVM technique is low, the number of FN is very high. It indicates that this model performs well in less likely recommended a movie on Netflix. It considers a limitation.

Furthermore, for the sentiment analysis section, by combining both positive and negative word cloud, it is clearly to see that most of the keywords are “movie”, “films”, “good”, “story”, “time” and so on. It indicates that all reviews contain the same keywords in both positive and negative review. The possible reason is these common keywords appeared in many reviews and some negative terms did not appear in the top 20 word.

In conclusion, this study successfully contributed to identify the champion model that can be used in predicting whether a movie is recommended to watch or less likely recommended to watch on Netflix. In the future, researchers can conduct more than 2 predictive modelling to identify the best techniques that can be used for movie recommendation system on Netflix. Next, the sentiments for all reviews are clearly to see that all users have a good attitude as they did not write many bad terms under a movie's review on Netflix. For the future recommendation to whom would like to study this topic, instead of identifying 1 term, the next study can focus on more than 1 term in the sentiment analysis.

7.0 REFERENCES

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